New L J Institute of Engineering & Technology

SEM: 1st – CSE/CSE (AIML)/DS

Sub: Mathematics-I

Sub Code: BE01R00041

Module 1: Basic Calculus

Weightage: 20%

Topics:

- **Chap: (5%)**

  Beta and Gamma functions and their properties.

❖ Chap : Beta and Gamma functions.

In this chapter, we will study two special functions, ***Gamma*** and ***Beta*** functions.

- The Beta function is also known as the Euler integral of the first kind and the Gamma function is known as the Euler integral of the second kind.

❖ Gamma Function:

The Gamma function is denoted and defines as

$$\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx \quad , \quad n > 0$$

➢ **Properties of Gamma Function:**

(1) $\Gamma(n+1) = n\Gamma n$

(2) $\Gamma(n+1) = n!$

(3) $\Gamma n = (n-1)!$

(4) $\Gamma(1/2) = \sqrt{\pi}$

Example 1: Show that $\Gamma(\mathrm{m}+1) = m\Gamma m$, where Γ is the Gamma function and m is a positive integer.

Solution: by definition of gamma function,

$$\Gamma m = \int_0^\infty e^{-x} x^{m-1} dx \quad , \quad m > 0$$

$$\Gamma(m+1) = \int_0^\infty e^{-x} x^{m} dx \quad , \quad m > 0$$

Integration by parts,

$$= [-e^{-x}\, x^m]_0^\infty - \int_0^\infty (-e^{-x}) m x^{m-1} dx$$

$$= 0 + m \int_0^\infty e^{-x}\, x^{m-1} dx$$

$$= 0 + m \int_0^\infty e^{-x}\, x^{m-1} dx$$

$$= m\Gamma m$$

Example 2: Show that $\Gamma(\mathrm{n}+1) = \mathrm{n}!$

Solution: We know that $\Gamma(\mathrm{n}+1) = n\Gamma n$……..(1)

$$\Gamma n = (n-1)\Gamma(n-1)$$

$$\Gamma(n-1) = (n-2)\Gamma(n-2)$$

$\Gamma(n-2) = (n-3)\Gamma(n-3)$ and so on

By repeated application of formula (1) we have

$$\Gamma(\mathrm{n}+1) = n(n-1)(n-2) \ldots 3.2.1\Gamma 1$$

But $\Gamma 1 = \int_0^\infty e^{-x}\, x^0\, dx = \int_0^\infty e^{-x}\, 1\, dx$

$$= \int_0^\infty e^{-x}\ dx$$

$$= |-e^{-x}|_0^\infty = -(e^{-\infty} - 1) = -(0 - 1) = 1$$

$$\therefore \Gamma(\mathrm{n}+1) = n(n-1)(n-2)\ldots 3.2.1 = \mathrm{n}!$$

Example 2: Define Gamma function and evaluate $\int_0^\infty e^{-x^2}\,dx$

Solution: let $x^2 = t \rightarrow x = t^{1/2}$

$$\therefore dx = \frac{1}{2}t^{-1/2}dt$$

When $x = 0 \rightarrow t = 0$ and When $x = \infty \rightarrow t = \infty$

$$\therefore \int_0^\infty e^{-x^2}\,dx = \int_0^\infty e^{-t}\ \frac{1}{2}t^{-1/2}\,dt$$

$$= \frac{1}{2}\int_0^\infty e^{-t}\ t^{\frac{1}{2}-1}\ dt$$

$$= \frac{1}{2}\Gamma(1/2)$$

$$= \frac{1}{2}\sqrt{\pi} \qquad \because \Gamma(1/2) = \sqrt{\pi}$$

Example 3: Prove that (i) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

Solution: by alternative form of gamma function

$$\Gamma n = 2\int_0^\infty e^{-x^2}x^{2n-1}dx \quad , \quad n > 0$$

$$\therefore \Gamma(1/2) = 2\int_0^\infty e^{-x^2}x^{2\left(\frac{1}{2}\right)-1}dx = 2\int_0^\infty e^{-x^2}dx$$

Now $\Gamma(1/2)\cdot\Gamma(1/2) = 2\int_0^\infty e^{-x^2}dx \cdot 2\int_0^\infty e^{-y^2}dy$

$$= 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\, dx\, dy$$

By changing into polar co-ordinates, $x = r\cos\theta,\ y = r\sin\theta$,we get

$$[\Gamma(1/2)]^2 = 4\int_{\theta=0}^{\theta=\frac{\pi}{2}} \int_{r=0}^{r=\infty} e^{-r^2}\, r\, dr\, d\theta$$

$$= 4\int_0^{\frac{\pi}{2}} d\theta \cdot \int_0^\infty \left(\frac{-1}{2}\right) e^{-r^2}(-2r)\, dr$$

$$= 4[\theta]_0^{\pi/2} \cdot \left(\frac{-1}{2}\right)\int_0^\infty e^{-r^2}(-2r)\, dr$$

$$= 4\left(\frac{\pi}{2}\right)\cdot\left(\frac{-1}{2}\right)\int_0^\infty e^{-r^2}(-2r)\, dr$$

$$= -\pi\left[e^{-r^2}\right]_0^\infty \qquad \because \int e^{f(x)} f'(x)dx = e^{f(x)}$$

$$= -\pi[e^{-\infty} - e^0] = -\pi[0-1] = \pi$$

$\therefore [\Gamma(1/2)]^2 = \pi$

$\therefore \Gamma\left(\mathbf{\frac{1}{2}}\right) = \sqrt{\boldsymbol{\pi}}$

Example 4 : Define Gamma function and Evaluate $\int_0^\infty x^6 e^{-2x}dx$.

Solution: let $2x = t \rightarrow x = \frac{t}{2}$

$$\therefore dx = \frac{dt}{2}$$

When $x = 0,\quad t = 0$ and When $x \rightarrow \infty,\quad t \rightarrow \infty$

$$\therefore \mathrm{I} = \int_0^\infty x^6 e^{-2x} dx = \int_0^\infty \left(\frac{t}{2}\right)^6 e^{-t} \frac{dt}{2}$$

$$= \frac{1}{128} \int_0^\infty e^{-t} \; t^6 \; dt$$

$$= \frac{1}{128} \int_0^\infty e^{-t} \; t^{7-1} \; dt$$

$$= \frac{1}{128} \Gamma(7)$$

$$= \frac{6!}{128} = \frac{45}{8}$$

Example 5 : Evaluate $\int_0^\infty 3^{-4x^2} dx$.

Solution: let $3^{-4x^2} = e^{-t}$ then taking log on both sides

$\therefore \; -4x^2 log3 = -t \; loge$

$\therefore \; 4x^2 log3 = t$

$\therefore \; x^2 = \frac{t}{4 \; log3}$

$\therefore \; x = \frac{\sqrt{t}}{2\sqrt{log3}}$

$$\therefore \; dx = \frac{1}{2\sqrt{log3}} \cdot \frac{1}{2\sqrt{t}} dt$$

When $x = 0$, $t = 0$ and When $x \to \infty$, $t \to \infty$

$$\therefore \mathrm{I} = \int_0^\infty 3^{-4x^2} dx = \int_0^\infty e^{-t} \cdot \frac{1}{2\sqrt{log3}} \cdot \frac{1}{2\sqrt{t}} dt$$

$$= \frac{1}{4\sqrt{log3}} \cdot \int_0^\infty e^{-t}\, t^{-1/2} \cdot dt$$

$$= \frac{1}{4\sqrt{log3}} \cdot \int_0^\infty e^{-t}\, t^{\frac{1}{2}-1} \cdot dt$$

$$= \frac{1}{4\sqrt{log3}} \cdot \Gamma(1/2) = \frac{\sqrt{\pi}}{4\sqrt{log3}} \cdot$$

Example 6: Prove that $\int_0^\infty \frac{x^c}{c^x} dx = \frac{\Gamma(c+1)}{(\log c)^{c+1}}$

Solution: Let $c^x = e^t \rightarrow x\, logc = t \rightarrow x = \frac{t}{logc}$

$$\therefore dx = \frac{dt}{\log c}$$

When $x = 0$, $t = 0$ and When $x \rightarrow \infty$, $t \rightarrow \infty$

$$\therefore \mathrm{I} = \int_0^\infty \frac{x^c}{c^x} dx = \int_0^\infty \left(\frac{t}{logc}\right)^c \cdot \frac{1}{e^t} \cdot \frac{1}{\log c}\, dt$$

$$= \frac{1}{(\log c)^{c+1}} \int_0^\infty e^{-t} \cdot t^c\, dt$$

$$= \frac{1}{(\log c)^{c+1}} \Gamma(\mathrm{c}+1)$$

$$= \frac{\Gamma(\mathrm{c}+1)}{(\log c)^{c+1}}$$

Example 7: Express $\int_0^1 x^m (1-x^n)^p\, dx$ in terms of Gamma function

Solution: Let $x^n = u \rightarrow x = u^{\frac{1}{n}}$

$$\therefore nx^{n-1} dx = du$$

$$\therefore dx = \frac{du}{nx^{n-1}} = \frac{du \cdot x}{nx^n} = \frac{du \cdot u^{\frac{1}{n}}}{nu}$$

When $x = 0$, $u = 0$ and When $x = 1$, $u = 1$

$$\therefore \mathrm{I} = \int_0^1 x^m (1 - x^n)^p dx = \int_0^1 u^{\frac{m}{n}} (1-u)^p \frac{u^{\frac{1}{n}}}{nu} du$$

$$= \frac{1}{n}\int_0^1 u^{\frac{m}{n}} \cdot u^{\frac{1}{n}} \cdot u^{-1}(1-u)^p \, du$$

$$= \frac{1}{n}\int_0^1 u^{\left(\frac{m+1}{n}\right)-1} \cdot (1-u)^{(p+1)-1} \, du$$

$$= \frac{1}{n} B\left(\frac{m+1}{n}, p+1\right)$$

$$= \frac{1}{n} \frac{\Gamma\left(\frac{m+1}{n}\right) \Gamma(p+1)}{\Gamma\left(\frac{m+1}{n} + p + 1\right)}$$

Example 8: Prove that $Gamma(n) = (n-1)Gamma(n-1)$.

Solution: by definition of gamma function,

$$\Gamma m = \int_0^\infty e^{-x} x^{m-1} dx \quad , \quad m > 0$$

$$\Gamma(m+1) = \int_0^\infty e^{-x} x^m dx \quad , \quad m > 0$$

Integration by parts,

$$= [-e^{-x}\, x^m]_0^\infty - \int_0^\infty (-e^{-x}) m x^{m-1} dx$$

$$= 0 + m\int_0^\infty e^{-x}\, x^{m-1} dx$$

$$= 0 + m\int_0^\infty e^{-x}\, x^{m-1} dx$$

$\therefore\ \Gamma(m+1) = m\Gamma m$ _____(1)

Now substitute $m = n - 1$ in (1)

$\therefore\ \Gamma(n - 1\ + 1) = n - 1\ \Gamma(n - 1)$

$\therefore\ \Gamma(n) = n - 1\ \Gamma(n - 1)$

❖ Beta Function:

The Beta function is denoted and defined as

$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx,\ \ m > 0, n > 0$$

➢ **Properties of Beta Function:**

(1) Symmetry $B(m,n) = B(n,m)$

(2) Relation between Beta and Gamma function

$$B(m,n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

(3) Trigonometric form of Beta function

$$B(m,n) = 2\int_0^{\frac{\pi}{2}} sin^{2m-1}x\ cos^{2n-1}x\ dx$$

(4) Beta function as a improper integral of first kind

$$B(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\ dx$$

Example 1: Prove that $\beta(m,n) = \beta(m,n+1) + \beta(m+1,n)$

Solution: **R.H.S** $= B(m,n+1) + B(m+1,n)$

$$= \frac{\Gamma m \Gamma(n+1)}{\Gamma(m+n+1)} + \frac{\Gamma(m+1)\Gamma n}{\Gamma(m+n+1)} \qquad \because\ B(m,n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

$$= \frac{\Gamma m\, n\Gamma n}{(m+n)\,\Gamma(m+n)} + \frac{m\Gamma m\ \ \Gamma n}{(m+n)\,\Gamma(m+n)} \qquad \because\ \Gamma(n+1) = n\Gamma n$$

$$= \frac{\Gamma m\, \Gamma n\, (m+n)}{(m+n)\,\Gamma(m+n)}$$

$$= \frac{\Gamma m\, \Gamma n}{\Gamma(m+n)}$$

$= B(m,n) =$ **L.H.S**

Example 2: Using the relationship between beta and gamma functions, simplify the expression $\beta(m,n)\,\beta(m+n,p)\,\beta(m+n+p,q)$.

Solution: $B(m,n)\,B(m+n,p)\,B(m+n+p,q)$

$$= \frac{\Gamma m\, \Gamma n}{\Gamma(m+n)} \cdot \frac{\Gamma(m+n)\Gamma p}{\Gamma(m+n+p)} \cdot \frac{\Gamma(m+n+p)\Gamma q}{\Gamma(m+n+p+q)}$$

$$= \frac{\Gamma m\, \Gamma n\, \Gamma p\, \Gamma q}{\Gamma(m+n+p+q)}$$

Example 3: **(i)** Find $\beta(5, 4)$.

Solution: Now, using the Beta – Gamma relation formula:

$$B(m,n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

Here $m = 5$ and $n = 4$

$$\therefore\ B(5\,,4) = \frac{\Gamma 5\ \ \Gamma 4}{\Gamma 9}$$

$$\therefore\ B(5\,,4) = \frac{4!\ \ 3!}{8!} \qquad \because\ \ \Gamma n = (n-1)!$$

$$\therefore\ B(5\,,4) = \frac{24\ \cdot 6}{40320} \qquad \because\ \ \Gamma n = (n-1)!$$

$$\therefore\ B(5\,,4) = \frac{144}{40320} = \frac{1}{280}$$

(ii) Find $\beta\left(\frac{9}{2},\frac{7}{2}\right)$

Solution: Using the Beta – Gamma relation formula:

$$B(m,n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

$$B\left(\frac{9}{2},\frac{7}{2}\right) = \frac{\Gamma(^9/_2)\Gamma(^7/_2)}{\Gamma(^9/_2 + {}^7/_2)}$$

$$= \frac{\Gamma(^7/_2 + 1)\,\Gamma(^7/_2)}{\Gamma 8}$$

$$= \frac{^7/_2\ \Gamma(^7/_2)\cdot\ \Gamma(^7/_2)}{7!}$$

$$= \frac{^7/_2\ [\Gamma(^7/_2)]^2}{7!}$$

$$= \frac{\left[\Gamma\left(^5/_2 + 1\right)\right]^2}{1440}$$

$$= \frac{\left[^5/_2\ \Gamma\left(^5/_2\right)\right]^2}{1440}$$

$$= \frac{\left[\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(1/2\right)\right]^2}{1440} \qquad repeated\ use\ of\ \Gamma(n+1) = n\Gamma n$$

$$= \frac{225}{92160}\pi$$

$$= \frac{5\pi}{2048}$$

(iii) Find the value $\beta\left(\frac{7}{2},\frac{5}{2}\right)$

Solution: Using the Beta – Gamma relation formula:

$$B(m,n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

$$B\left(\frac{7}{2},\frac{5}{2}\right) = \frac{\Gamma\left(7/2\right)\Gamma\left(5/2\right)}{\Gamma\left(7/2 + 5/2\right)}$$

$$= \frac{\Gamma\left(5/2 + 1\right)\Gamma\left(5/2\right)}{\Gamma 6}$$

$$= \frac{5/2\ \Gamma\left(5/2\right)\cdot\Gamma\left(5/2\right)}{5!}$$

$$= \frac{5/2\ \left[\Gamma\left(5/2\right)\right]^2}{120}$$

$$= \frac{\left[\Gamma\left(3/2 + 1\right)\right]^2}{48}$$

$$= \frac{\left[3/2\ \Gamma\left(3/2\right)\right]^2}{48}$$

$$= \frac{\left[\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(1/2\right)\right]^2}{48} \qquad repeated\ use\ of\ \Gamma(n+1) = n\Gamma n$$

$$= \frac{9\pi}{16\times 48}$$

$$= \frac{3\pi}{256}$$

(iv) Find the value of $\beta(2,3)$

Solution: Using the Beta – Gamma relation formula:

$$B(m,n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

$$\therefore\ B(2\,,3) = \frac{\Gamma 2\ \ \Gamma 3}{\Gamma 5}$$

$$\therefore\ B(2\,,3) = \frac{1!\ \ 2!}{4!} \qquad \because\ \ \Gamma n = (n-1)!$$

$$\therefore\ B(2\,,3) = \frac{1\ \cdot\ 2}{24} = \frac{1}{12}$$

Example 4: Evaluate $\int_0^1 x^4\left(1-\sqrt{x}\right)^5 dx$.

Solution: Let $\sqrt{x} = t \rightarrow x = t^2$

$$dx = 2t\ dt$$

When $x = 0,\qquad t = 0$

When $x = 1,\qquad t = 1$

$$\therefore \int_0^1 x^4\left(1-\sqrt{x}\right)^5 dx = \int_0^1 t^8\,(1-t)^5\ 2t\ dt$$

$$= 2\int_0^1 t^9\,(1-t)^5\ dt$$

$$= \ 2\int_0^1 t^{\mathbf{10-1}}\,(1-t)^{\mathbf{6-1}}\ dt$$

$$= 2\,B(10\,,6)$$

$$= 2\,\frac{\Gamma 10\ \ \Gamma 6}{\Gamma 16}$$

$$= 2\,\frac{9!\ \ 5!}{15!}$$

$$= \frac{1}{15015}$$

Example 5: Evaluate $\int_0^1 x^3\left(1-\sqrt{x}\right)^5 dx$.

Solution: Let $\sqrt{x} = t \rightarrow x = t^2$

$$dx = 2t\ dt$$

When $x = 0,\qquad t = 0$

When $x = 1,\qquad t = 1$

$$\therefore \int_0^1 x^3\left(1-\sqrt{x}\right)^5 dx = \int_0^1 t^6\,(1-t)^5\ 2t\ dt$$

$$= 2\int_0^1 t^7\,(1-t)^5\ dt$$

$$= \ 2\int_0^1 t^{\mathbf{8-1}}\,(1-t)^{\mathbf{6-1}}\ dt$$

$$= 2\,B(8\,,6)$$

$$= 2\,\frac{\Gamma 8\ \ \Gamma 6}{\Gamma 14}$$

$$= 2\,\frac{7!\ \ 5!}{13!}$$

$$= \frac{1}{5148}$$

Example 6 : Prove that $\int_0^2 x^4(8-x^3)^{-1/3}dx = \frac{16}{3}\beta(\frac{5}{3},\frac{2}{3})$

Solution: L.H.S = $\int_0^2 x^4(8-x^3)^{-1/3}dx$

Let $x^3 = 8t$

$\therefore x = (8t)^{1/3} = 2t^{1/3}$

$\therefore dx = 2\cdot\frac{1}{3}\,t^{\frac{-2}{3}}\,dt$

When $x = 0,$ $t = 0$

When $x = 2,$ $t = 1$

$$\therefore \int_0^2 x^4(8-x^3)^{-1/3}dx = \int_0^1 \left(2t^{1/3}\right)^4\cdot(8-8t)^{\frac{-1}{3}}\cdot\cdot\frac{2}{3}\,t^{\frac{-2}{3}}\,dt$$

$$= \int_0^1 16\,t^{4/3}\cdot(8)^{\frac{-1}{3}}(1-t)^{\frac{-1}{3}}\cdot\frac{2}{3}\,t^{\frac{-2}{3}}\,dt$$

$$= \frac{16}{3}\int_0^1 t^{\frac{2}{3}}\cdot(1-t)^{\frac{-1}{3}}\,dt$$

$$= \frac{16}{3}\int_0^1 t^{\frac{5}{3}-1}\cdot(1-t)^{\frac{2}{3}-1}\,dt$$

$$= \frac{16}{3}\,B\left(\frac{5}{3},\frac{2}{3}\right) = \textbf{R.H.S}$$

Example 7 : Evaluate $\int_3^7 \sqrt[4]{(x-3)(7-x)}\, dx$ using the concept of Gamma and Beta function

Solution: Let $x - 3 = t \rightarrow dx = dt$

When $x = 3, \quad t = 0$

When $x = 7, \quad t = 4$

$$\therefore I = \int_3^7 \sqrt[4]{(x-3)(7-x)}\, dx = \int_3^7 (x-3)^{1/4}(7-x)^{1/4}\, dx$$

$$= \int_0^4 t^{1/4}\,(4-t)^{1/4}\, dt$$

Again let $t = 4u \rightarrow dt = 4du$

When $t = 0, \quad u = 0$ and When $t = 4, \quad u = 1$

$$\therefore I = \int_0^1 (4u)^{1/4}\,(4-4u)^{1/4}\;\, 4du$$

$$= \int_0^1 4^{1/4}\cdot(u)^{1/4}\;\, 4^{1/4}\,(1-u)^{1/4}\;\, 4du$$

$$= 8\int_0^1 (u)^{1/4}\;\,(1-u)^{1/4}\, du$$

$$= 8\int_0^1 (u)^{\frac{5}{4}-1}\;\,(1-u)^{\frac{5}{4}-1}\, du$$

$$= 8\, B\left(\frac{5}{4},\frac{5}{4}\right)$$

$$= 8\,\frac{\Gamma\left(5/4\right)\Gamma\left(5/4\right)}{\Gamma\left(5/2\right)}$$

$$= 8\ \frac{\left[\Gamma\left(5/4\right)\right]^2}{\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(1/2\right)}$$

$$= 8\ \frac{\left[\Gamma\left(1/4\ +1\right)\right]^2}{\frac{3}{4}\sqrt{\pi}}$$

$$= \frac{2}{3}\frac{\left[\Gamma\left(1/4\right)\right]^2}{\sqrt{\pi}}$$

Example 8 : Using the Beta and Gamma functions evaluate the integral $\int_{-1}^{1}(1-x^2)^n dx$, where n is a positive integer.

Solution: Here $(1-x^2)^n$ is an even function because

$$f(-x) = [1-(-x)^2]^n = (1-x^2)^n = f(x)$$

$$\therefore I = \int_{-1}^{1}(1-x^2)^n dx = 2\int_{0}^{1}(1-x^2)^n\, dx$$

Let $x^2 = t \rightarrow 2x\, dx = dt$

$$\therefore 2\, dx = \frac{1}{x}dt = \frac{1}{\sqrt{t}}dt$$

When $x = 0,\quad t = 0$ and When $x = 1,\quad t = 1$

$$\therefore I = \int_0^1 (1-t)^n \frac{1}{\sqrt{t}}dt$$

$$= \int_0^1 t^{\frac{-1}{2}}(1-t)^n\, dt$$

$$= \int_0^1 t^{\frac{1}{2}-1}(1-t)^{(n+1)-1}\, dt$$

$$= B\left(\frac{1}{2}, n+1\right)$$

$$= \frac{\Gamma(1/2)\Gamma(n+1)}{\Gamma(1/2 + n + 1)}$$

$$\therefore I = \frac{\sqrt{\pi}\, n!}{\Gamma(n + 3/2)}$$

Example 9: Investigate convergence of the following integrals: $\int_0^\infty \frac{x^{10}\,(1+x^5)}{(1+x)^{27}}\,dx$

Solution: Let $I = \int_0^\infty \frac{x^{10}\,(1+x^5)}{(1+x)^{27}}\,dx$

$$= \int_0^\infty \frac{x^{10}}{(1+x)^{27}}\,dx + \int_0^\infty \frac{x^{15}}{(1+x)^{27}}\,dx$$

$$= \int_0^\infty \frac{x^{11-1}}{(1+x)^{11+16}}\,dx + \int_0^\infty \frac{x^{16-1}}{(1+x)^{16+11}}\,dx$$

$$= B(11,16) + B(16,11) \qquad \because B(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx$$

$$= 2\,B(11,16) \qquad \because B(m,n) = B(n,m)$$

$$= 2\,\frac{\Gamma(11)\,\Gamma(16)}{\Gamma(27)}$$

$$= 2\,\frac{10!\;\;15!}{26!}$$

Which is a finite quantity

$\therefore$ The given improper integral converges.

Example 10: State the Duplication formula. Show that $\int_0^\infty \sqrt{x}\, e^{-x^2}\, dx \int_0^\infty \frac{e^{-x^2}}{\sqrt{x}} dx = \frac{\pi}{2\sqrt{2}}$

Solution: The Duplication formula for gamma function is

$$\Gamma(\mathrm{n})\Gamma\left(\mathrm{n}+\frac{1}{2}\right) = \frac{\sqrt{\pi}\ \Gamma(2\mathrm{n})}{2^{2n-1}}$$

Let $\mathrm{I}_1 = \int_0^\infty \sqrt{x}\, e^{-x^2}\, dx$

Let $x^2 = t \to x = t^{\frac{1}{2}}$

$\therefore 2x\, dx = dt$

$\therefore dx = \frac{dt}{2x} = \frac{dt}{2t^{\frac{1}{2}}}$

When $x = 0$, $t = 0$ and When $x \to \infty$, $t \to \infty$

$$\therefore \mathrm{I}_1 = \int_0^\infty \sqrt{x}\, e^{-x^2}\, dx = \int_0^\infty t^{\frac{1}{4}}\, e^{-t} \frac{dt}{2t^{\frac{1}{2}}}$$

$$= \frac{1}{2}\int_0^\infty t^{\frac{-1}{4}} e^{-t}\, dt$$

$$= \frac{1}{2}\int_0^\infty e^{-t}\ \ t^{\frac{3}{4}-1}\ dt$$

$$= \frac{1}{2}\Gamma\left(\frac{3}{4}\right) \qquad \because\ \Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$$

Let $\mathrm{I}_2 = \int_0^\infty \frac{e^{-x^2}}{\sqrt{x}} dx$

Let $x^2 = t \to x = t^{\frac{1}{2}}$

$\therefore 2x\, dx = dt \qquad \therefore dx = \frac{dt}{2x} = \frac{dt}{2t^{\frac{1}{2}}}$

When $x = 0$, $t = 0$ and When $x \to \infty$, $t \to \infty$

$$I_2 = \int_0^\infty \frac{e^{-x^2}}{\sqrt{x}}\,dx = \int_0^\infty x^{\frac{-1}{2}} e^{-x^2}\,dx = \int_0^\infty t^{\frac{-1}{4}}\, e^{-t} \frac{dt}{2t^{\frac{1}{2}}}$$

$$= \frac{1}{2}\int_0^\infty t^{\frac{-3}{4}}\, e^{-t}\,dt$$

$$= \frac{1}{2}\int_0^\infty e^{-t}\, t^{\frac{1}{4}-1}\,dt$$

$$= \frac{1}{2}\Gamma\left(\frac{1}{4}\right) \qquad \because\ \Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$$

Now $I_1 \times I_2 = \int_0^\infty \sqrt{x}\, e^{-x^2}\,dx \times \int_0^\infty \frac{e^{-x^2}}{\sqrt{x}}\,dx$

$$= \frac{1}{4}\left[\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right)\right] \ldots\ldots\ldots\ldots(1)$$

$$= \frac{1}{4}\left[\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{1}{4}+\frac{1}{2}\right)\right]$$

$$= \frac{1}{4}\left[\frac{\sqrt{\pi}\,\Gamma\left(\frac{1}{2}\right)}{2^{\left(2\times\frac{1}{4}\right)-1}}\right] \qquad \textit{using Dupication formula } n = \frac{1}{4}$$

$$= \frac{1}{4}\left[\frac{\sqrt{\pi}\,\sqrt{\pi}}{2^{\frac{-1}{2}}}\right]$$

$$= \frac{1}{4}\left[\pi \times \sqrt{2}\right]$$

$$= \frac{\pi}{2\sqrt{2}}$$